

Haoxiang Wan

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Education

University of Houston, Houston, TX

Aug. 2025 – Present

Ph.D. in Electrical and Computer Engineering

Advisor: [Prof. Xingpeng Li](#), Renewable Power Grid (RPG) Laboratory

University of Illinois at Urbana-Champaign, Urbana, IL

May 2025

Master of Engineering in Energy Systems

The University of Sheffield, Sheffield, UK

June 2024

Bachelor of Engineering in Electrical and Electronic Engineering, *First Class Honours*

Research Interests

Power system optimization; data center load integration and flexibility; generation and transmission expansion planning; hosting capacity assessment; GPU-accelerated large-scale optimization.

Research Experience

Graduate Research Assistant

Aug. 2025 – Present

Renewable Power Grid (RPG) Laboratory, University of Houston, Houston, TX

Advisor: [Prof. Xingpeng Li](#)

- Built security-constrained unit commitment (SCUC) models that shift data center load across buses and hours, reducing transmission congestion, renewable curtailment, and operating cost.
- Developed a co-optimized generation and transmission expansion planning model with embedded unit commitment on IEEE RTS-24, showing that 500 MW of data center load is served by new generation instead of new lines and raises total planning cost by 62%.
- Compared four security models, from OPF to N-1 SCUC, for bus-level data center hosting capacity, quantifying how much simplified models overstate the load a bus can accommodate.

Research Aide (Summer Internship)

May 2026 – Aug. 2026

Argonne National Laboratory, Lemont, IL

Supervisor: [Dr. Jonghwan Kwon](#), Energy Systems and Infrastructure Analysis Division

- Built a Monte Carlo cross-stress framework for generation expansion planning under uncertain data center deployment, evaluating 30 candidate portfolios against 30 deployment scenarios on a 254-bus Texas 2030 system.
- Solved 1,161 full-year (8,760-hour) linear programs on GPUs with NVIDIA cuOpt/PDLP in Argonne's [A-LEAF](#) framework.

Publications

Published and Accepted

- **H. Wan** and X. Li, “Data Center Spatio-Temporal Load Flexibility in Security-Constrained Unit Commitment for Enhanced Grid Efficiency and Reliability,” to appear in *Proc. IEEE Industry Applications Society (IAS) Annual Meeting*, Vancouver, BC, Canada, Oct. 2026. [[arXiv](#)]
- **H. Wan**, L. Fang, and X. Li, “Grid Operational Benefit Analysis of Data Center Spatial Flexibility: Congestion Relief, Renewable Energy Curtailment Reduction, and Cost Saving,” in *Proc. IEEE Power & Energy Society General Meeting (PESGM)*, Montreal, QC, Canada, Jul. 2026. [[arXiv](#)]

Under Review

- **H. Wan**, L. Fang, J. Kwon, J. Silva-Rodriguez, J. Yang, J. Yellajosula, F. Jiang, and X. Li, “Impact of Security Modeling Fidelity on Hyperscale Data Center Hosting Capacity Assessment,” submitted to *IEEE PES Grid Edge Technologies Conference & Exposition*, 2027.
- **H. Wan**, J. Kwon, and X. Li, “Impact of Data Center Load on Co-Optimized Generation and Transmission Expansion Planning with Embedded Unit Commitment,” submitted to *IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, 2027.
- L. Fang, **H. Wan**, A. Srivastava, and X. Li, “Service-Aware Hosting Capacity Assessment of Edge Data Centers

with Workload Flexibility,” submitted to *IEEE PES Grid Edge Technologies Conference & Exposition*, 2027.

In Preparation

- **H. Wan**, J. Kwon, S. Cho, S. B. Sharma, T. Levin, and X. Li, “Risk-Aware Selection of Generation Expansion Portfolios under Uncertain Data-Center Deployment: A Full-Year Cross-Stress Framework,” in preparation for *Applied Energy*, 2026.
- **H. Wan**, I. Castillo, and X. Li, “Beyond Unserved Energy: Modeled Reserve Scarcity Falls but Transmission Congestion Persists Across a Wind–Solar–Storage Sequence,” in preparation for *IEEE Gulf Coast Conference on Energy, Communications and Automation (GECA)*, 2027.

Teaching Experience

Graduate Teaching Assistant

Jan. 2026 – May 2026

ECE 5380 Power Electronics & Electric Drives, University of Houston

Instructor: [Dr. Hao Huang](#)

- Graded homework and examinations and provided written feedback for a synchronous online graduate course on power semiconductor devices, converters and inverters, and motor drives.

Graduate Teaching Assistant

Aug. 2025 – Dec. 2025

ECE 5388 Renewable Energy Technology, University of Houston

Instructor: [Dr. Hao Huang](#)

- Assisted with instruction and assessment for a course on solar, wind, and grid integration of renewable resources.

Technical Skills

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- **Optimization & Modeling:** Pyomo, JuMP, AMPL, Gurobi, CPLEX, NVIDIA cuOpt/PDLP, MadNLP
 - **Programming Languages:** Python, Julia, MATLAB, C, C++
 - **Power System Tools:** MATLAB/Simulink, LTspice
 - **Machine Learning:** PyTorch, scikit-learn, pandas, NumPy
 - **Computing:** Linux/HPC (Slurm, PBS), GPU computing (CUDA), Git
 - **Languages:** English (fluent), Mandarin Chinese (native)

Honors and Awards

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- Engineering Excellence Scholarship, The University of Sheffield *2021–2022, 2022–2023*
 - NCUK Sheffield Scholarship, The University of Sheffield *2021–2022, 2022–2023*

Professional Affiliations

Student Member, IEEE Power & Energy Society (PES) and IEEE Industry Applications Society (IAS)